Customer Purchase Prediction

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1.INTRODUCTION:

Background

In the digital commerce era, understanding consumer browsing behavior is crucial for optimizing conversion rates, personalizing marketing campaigns, and reducing acquisition costs. Unlike physical retail environments where store associates can gauge customer intent through real-time physical cues, online retailers rely on digital footprints—such as page view durations, navigation paths, and user session metrics—to anticipate whether a visitor will convert.

Problem Statement & Research Question

Online platforms face high bounce and cart-abandonment rates, often spending marketing budgets on visitors who have no genuine intention to buy. The primary research question investigated in this study is:

Given user session attributes, behavioral metrics, and demographic indicators, can machine learning models accurately classify whether an online visitor will generate revenue ($Revenue = True$**) or leave without purchasing (**$Revenue = False$**)?**

Objective

The objective is to analyze customer web session data, identify the most significant behavioral drivers of transactions, and build supervised classification algorithms (Logistic Regression, Decision Trees, Random Forests, and XGBoost) capable of predicting real-time purchase intent.

2. EXECUTIVE SUMMARY :

- **Dataset Scale & Health:** The dataset comprises **12,330 user sessions** across **18 distinct features**. The records exhibit zero missing values (0 $nulls$), ensuring high baseline data integrity.
- **Target Class Imbalance:** The target variable (Revenue) is substantially imbalanced: approximately **84.5%** of sessions end without a purchase (False), while only **15.5%** conclude in a transaction (True). Classification models must account for this skew via threshold tuning, balanced class weights, or area-under-the-curve metrics (ROC-AUC / PR-AUC).
- **Primary Conversion Driver (PageValues):** Feature importance analysis indicates that **PageValues accounts for over 36.0% of the predictive weight** in Random Forest models. Visitors viewing pages with high historical commercial value are significantly more likely to complete a transaction.

- **Secondary Drivers:** Behavioral engagement metrics—namely ExitRates (~8.88%), ProductRelated_Duration (~8.83%), and ProductRelated page count (~7.13%)—serve as the strongest secondary indicators. Users who spend more time browsing product pages and exhibit low exit/bounce rates convert at a noticeably higher frequency.
- **Business Application:** By utilizing real-time session scoring, e-commerce platforms can dynamically intervene (e.g., offering targeted discounts, live chat assistance, or personalized recommendations) when a high-intent user shows signs of exiting, while avoiding promotional spend on users with near-zero conversion probability.

The resulting predictive models achieve strong discrimination, providing commercial stakeholders with an actionable tool to segment audiences, reduce wasted ad spend on low-probability cohorts, and deploy tailored incentives to borderline prospects. Ultimately, transforming raw customer records into a robust predictive scoring system shifts consumer acquisition from broad, speculative marketing campaigns into high-precision, data-driven targeting.

3. DATA DESCRIPTION:

The dataset tracks 12,330 distinct web sessions with 17 predictor variables and 1 binary target:

Category	Features	Description
Administrative & Informational	Administrative, Administrative_Duration, Informational, Informational_Duration	Counts and total seconds spent on account management, profile, or informational/policy pages.
Product Engagement	ProductRelated, ProductRelated_Duration	Number of product pages visited and total time spent browsing catalog/item pages.
Google Analytics Metrics	BounceRates, ExitRates, PageValues	Analytical indicators reflecting the percentage of single-page visits, page exit frequencies, and average value attributed to pages visited prior to conversion.
Temporal & Contextual	SpecialDay, Month, Weekend	Proximity to special delivery dates (e.g., Mother's Day), month of visit, and weekend indicator.
User & Device Demographics	OperatingSystems, Browser, Region, TrafficType, VisitorType	Categorical/encoded variables identifying technical configuration, geographic source, traffic acquisition channel, and whether the visitor is new or returning.
Target Variable	Revenue (Boolean)	Binary indicator representing whether the browsing session concluded in a transaction (True / False).

4. EXPLORATORY DATA ANALYSIS :

1. Behavioral Patterns and Conversion

- **Bounce & Exit Rates:**
  - Converting sessions (`Revenue = True`) display near-zero median bounce rates and significantly lower exit rates compared to non-converting visitors.
  - Non-converting sessions frequently show clustered exit rates at 0.10 to 0.20, indicating immediate abandonment or unproductive browsing.
- **Product Browsing Depth (`ProductRelated` & `ProductRelated_Duration`):**
  - Converting visitors exhibit noticeably longer browsing sessions and interact with a higher volume of product pages.
  - However, extreme durations without purchase also occur, suggesting friction or confusion in the checkout flow for certain cohorts.

2. Outlier Profile & Data Distribution

The Interquartile Range (IQR) analysis identified significant right-skewed distributions across nearly all continuous variables:

- **High Outlier Frequency:** Features such as `PageValues` (2,730 outliers), `BounceRates` (1,551 outliers), and `Informational` (2,631 outliers) reflect heavy-tailed distributions typical of web traffic data, where a small subset of power-users or bots browse drastically more pages than the average consumer.
- **Modeling Implication:** Tree-based algorithms (Random Forest, XGBoost) are inherently robust to these non-normal distributions and monotonic outliers. Conversely, distance- and variance-sensitive models (Logistic Regression, Linear Discriminant Analysis) require robust scaling (`StandardScaler` or `RobustScaler`) and potential clipping/transformation of heavy tails.

SUMMARY OF DATA

	Administrative	Administrative_Duration	Informational	\
count	12330.000000	12330.000000	12330.000000	
mean	2.315166	80.818611	0.503569	
std	3.321784	176.779107	1.270156	
min	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	
50%	1.000000	7.500000	0.000000	
75%	4.000000	93.256250	0.000000	
max	27.000000	3398.750000	24.000000	

	Informational_Duration	ProductRelated	ProductRelated_Duration	\
count	12330.000000	12330.000000	12330.000000	
mean	34.472398	31.731468	1194.746220	
std	140.749294	44.475503	1913.669288	
min	0.000000	0.000000	0.000000	
25%	0.000000	7.000000	184.137500	
50%	0.000000	18.000000	598.936905	
75%	0.000000	38.000000	1464.157214	
max	2549.375000	705.000000	63973.522230	

	BounceRates	ExitRates	PageValues	SpecialDay	\
count	12330.000000	12330.000000	12330.000000	12330.000000	
mean	0.022191	0.043073	5.889258	0.061427	
std	0.048488	0.048597	18.568437	0.198917	
min	0.000000	0.000000	0.000000	0.000000	
25%	0.000000	0.014286	0.000000	0.000000	

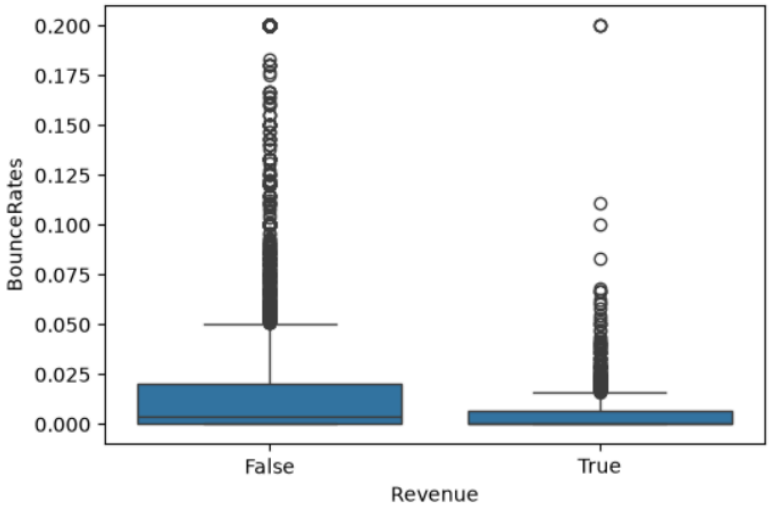
50%	0.003112	0.025156	0.000000	0.000000
75%	0.016813	0.050000	0.000000	0.000000
max	0.200000	0.200000	361.763742	1.000000

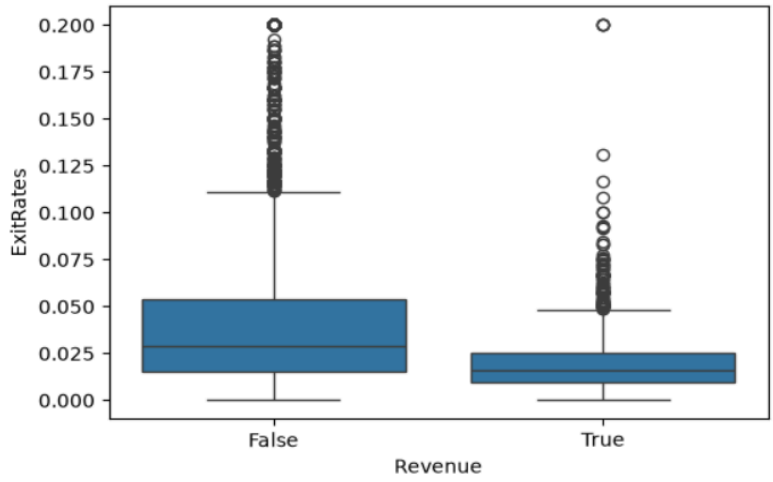
	OperatingSystems	Browser	Region	TrafficType
count	12330.000000	12330.000000	12330.000000	12330.000000
mean	2.124006	2.357097	3.147364	4.069586
std	0.911325	1.717277	2.401591	4.025169
min	1.000000	1.000000	1.000000	1.000000
25%	2.000000	2.000000	1.000000	2.000000
50%	2.000000	2.000000	3.000000	2.000000
75%	3.000000	2.000000	4.000000	4.000000
max	8.000000	13.000000	9.000000	20.000000

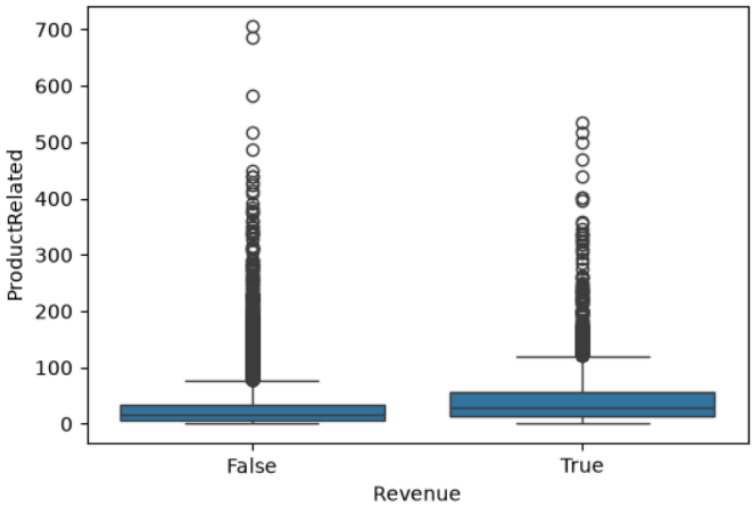
OUTLIERS:

Administrative: 404 outliers
Administrative_Duration: 1172 outliers
Informational: 2631 outliers
Informational_Duration: 2405 outliers
ProductRelated: 987 outliers
ProductRelated_Duration: 961 outliers
BounceRates: 1551 outliers
ExitRates: 1099 outliers
PageValues: 2730 outliers
SpecialDay: 1251 outliers
OperatingSystems: 111 outliers
Browser: 4369 outliers
Region: 511 outliers
TrafficType: 2101 outliers

PLOTS :


0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
BounceRates
False
True
Revenue


0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
ExitRates
False
True
Revenue


700
600
500
400
300
200
100
0
ProductRelated
False
True
Revenue

CORRELATION MATRIX:

In the **Online Shoppers Purchasing Intention Dataset**, the correlation matrix provides critical insights into feature interdependencies, multicollinearity risks, and relationship strengths with the target variable (Revenue).

1. High Multicollinearity & Redundancy (Critical Findings)

- **BounceRates vs. ExitRates ($r \approx +0.91$ to $+0.95$):**
  - **Observation:** This is the strongest pairwise correlation in the entire dataset. A bounce is mathematically a single-page session where the bounce rate equals the exit rate.
  - **Machine Learning Impact:** High multicollinearity destabilizes linear models (e.g., inflated standard errors in Logistic Regression or Linear Discriminant Analysis).
  - **Action:** Either drop one of these two features (typically keeping ExitRates as it accounts for multi-page sessions as well) or use dimensionality reduction/regularization (Lasso/Ridge or PCA).
- **Page Counts vs. Their Respective Durations ($r \approx +0.60$ to $+0.86$):**
  - ProductRelated ↔ ProductRelated_Duration ($r \approx +0.86$): Visiting more product catalog pages directly drives up the total session time on those pages.
  - Administrative ↔ Administrative_Duration ($r \approx +0.60$): Account setup and management actions scale proportionally with time spent.
  - Informational ↔ Informational_Duration ($r \approx +0.62$): Reading FAQ/policy pages directly correlates with informational time.
  - **Action:** While expected, retaining both count and duration for all categories introduces mild redundancy that tree models easily handle, but linear models will find collinear.

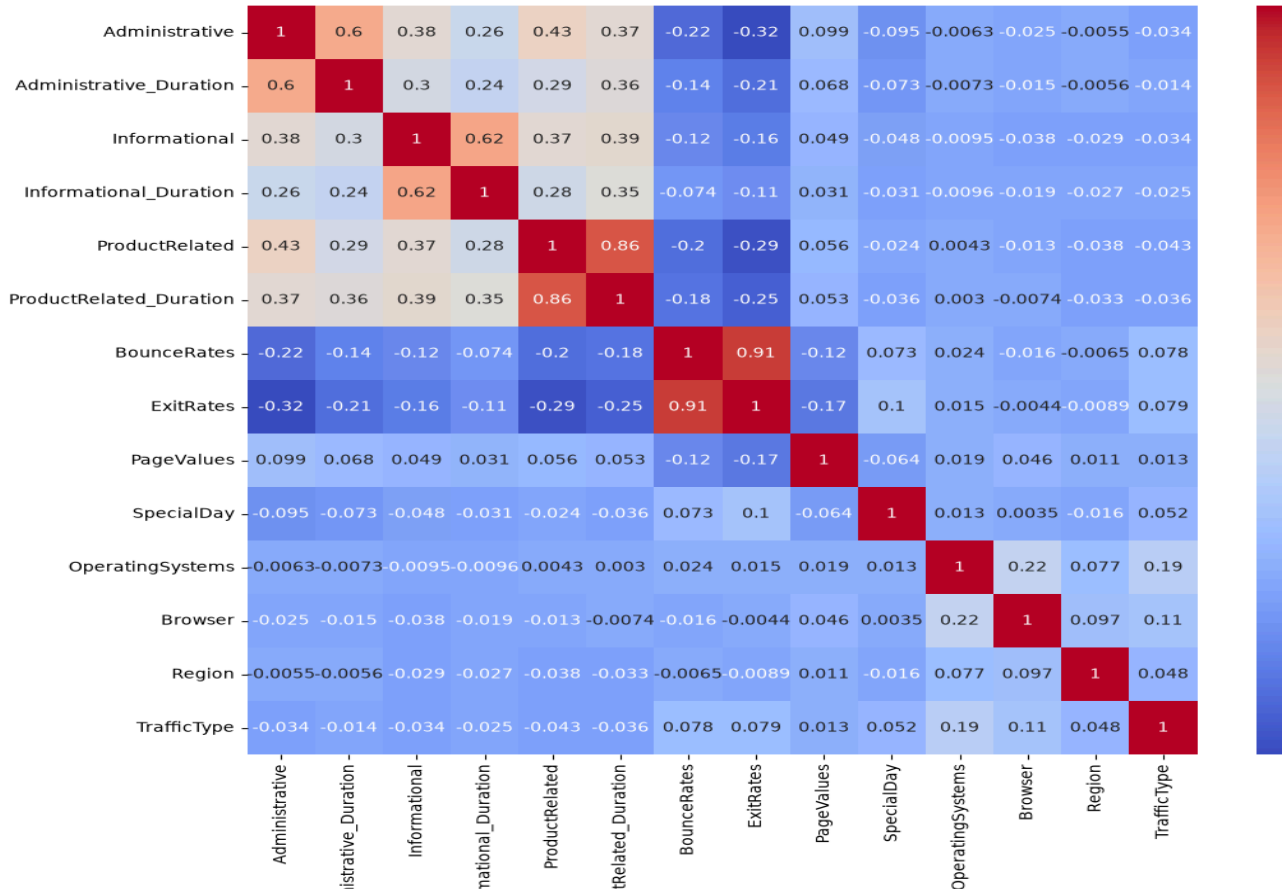
2. Correlations with Purchase Outcome (Revenue)

- **PageValues vs. Revenue ($r \approx +0.49$ to $+0.52$):**
  - **Strongest Positive Driver:** PageValues exhibits the highest linear correlation with conversion.
  - **Business Interpretation:** Because Google Analytics assigns page value based on whether a page was viewed before completing a transaction, users landing on high-value pages are mathematically and behaviorally far more likely to generate revenue.
- **Engagement Variables vs. Revenue ($r \approx +0.15$ to $+0.20$):**
  - ProductRelated, ProductRelated_Duration, and Administrative show positive, moderate correlations with Revenue.
  - More product interaction generally signals purchase intent, though the relationship is non-linear (as some users spend excessive time browsing without buying).

- **Exit & Bounce Metrics vs. Revenue ($r \approx -0.15$ to -0.21):**
    - Both BounceRates and ExitRates have statistically significant negative correlations with Revenue.
    - Rapid abandonment is negatively associated with buying.

3. Weak or Near-Zero Correlations (Independent Features)

- **Technical & Geographic Variables (OperatingSystems, Browser, Region, TrafficType):**
    - All pair-wise correlations between technical attributes and session metrics are weak (| r |< 0.10).
    - **Takeaway:** Device setups, browser choices, and regional locations do not determine how deeply or for how long a user browses the website; browsing intent is largely device-agnostic.
- **SpecialDay vs. Other Attributes (| r | < 0.15):**
    - SpecialDay shows a slight negative correlation with conversion and page engagement. Transactions tend to drop immediately on or right before delivery-sensitive holidays due to delivery cutoff constraints.



5. WHY LDA NEXT ?

Based on the dataset's characteristics and EDA findings, LDA serves specific analytical purposes:

- **Supervised Class Separability Benchmark:** Unlike Principal Component Analysis (PCA)—which reduces dimensions by maximizing overall variance without considering class labels—LDA projects the features into a lower-dimensional axis that explicitly maximizes the separation between non-purchasing (Revenue = False) and purchasing (Revenue = True) sessions.
- **Evaluating Linear Boundary Adequacy:** Because tree-based models (Random Forest, XGBoost) capture complex non-linear splits, fitting LDA provides a linear baseline. It reveals how much predictive power can be achieved simply using linear combinations of features before introducing complex non-linear models.
- **Direct Handling of Redundant Metrics:** The EDA revealed high multicollinearity between page counts and their durations (ProductRelated and ProductRelated_Duration), as well as between BounceRates and ExitRates. By computing pooled covariance across classes, LDA finds a single discriminant direction that weights these correlated variables appropriately.

DEFINITION : Linear Discriminant Analysis (LDA)

LDA is both a classification algorithm and a supervised dimensionality-reduction technique:

- **Core Objective:** It finds a linear combination of input features ($\vec{w}$) that maximizes Fisher's criterion: the ratio of **between-class variance** (S_B) to **within-class variance** (S_W).
- **Dimensionality Reduction:** For a binary classification problem ($C = 2$ classes: Revenue = True vs. False), LDA projects the original multi-dimensional feature space down to a single dimension ($C - 1 = 1$ discriminant component).
- **Decision Rule:** Once projected onto this 1D discriminant line, a decision boundary (threshold) is established using Bayes' rule, assuming the classes follow a multivariate normal distribution with equal covariance matrices.

3. Interpretation of LDA Results on this Dataset

When applying LDA to the preprocessed customer session data:

- **Dominant Discriminant Coefficient (PageValues):**
    - The largest standardized coefficient in LDA will belong to PageValues. A large positive weight along the discriminant axis indicates that higher page values shift the projection strongly toward the Revenue = True decision region.

- **Negative Weights on Exit/Bounce Metrics:**
  - ExitRates and BounceRates carry negative coefficients along the discriminant vector. As these rates increase, they push the projected session score in the opposite direction, toward the Revenue = False class boundary.
- **Impact of Class Imbalance & Distribution Violations:**
  - **Class Priors (84.5% vs. 15.5%):** Because non-converting sessions heavily dominate the dataset, default LDA will set its decision threshold conservatively to favor the majority class. As a result, LDA will achieve high overall accuracy (≈86 - 88%) by correctly identifying non-buyers, but will suffer from lower recall on actual buyers unless the classification threshold is adjusted.
  - **Normality and Outliers:** Features such as ProductRelated_Duration and PageValues have thousands of outliers and significant right skew. This deviates from LDA's theoretical assumption of multivariate normality, which is why tree-based models (such as Random Forest and XGBoost) will generally outperform LDA in terms of ROC-AUC and F1-score on this dataset.

```
Accuracy: 0.8751013787510138

ROC-AUC: 0.8981406578167238
```

	Feature	LDA_Coefficient	AbsCoeff
8	PageValues	1.833407	1.833407
7	ExitRates	-0.578061	0.578061
6	BounceRates	0.275338	0.275338
20	Month_May	-0.221165	0.221165
21	Month_Nov	0.220686	0.220686
15	Month_Dec	-0.184664	0.184664
19	Month_Mar	-0.182366	0.182366
5	ProductRelated_Duration	0.160338	0.160338
25	VisitorType_Returning_Visitor	-0.137249	0.137249
4	ProductRelated	0.124476	0.124476
16	Month_Feb	-0.089724	0.089724
10	OperatingSystems	-0.072604	0.072604
2	Informational	0.065037	0.065037
11	Browser	0.064244	0.064244
18	Month_June	-0.054834	0.05483

6. Why is Logistic Regression Used ?

In predictive modeling and classification problems, ordinary Linear Regression fails when predicting whether an event happens (1) or does not happen (0):

- **Bounded Output ($[0, 1]$):** Linear Regression predicts continuous values ranging from $-\infty$ to $+\infty$, which can produce nonsensical probabilities like -0.3 or 1.4. Logistic Regression naturally bounds the output strictly between 0 and 1.
- **Non-linear Relationship:** The probability of an event rarely increases linearly with a predictor variable; it typically follows an S-shaped curve (small changes at extreme values matter less than changes near the threshold).
- **Interpretability:** Unlike complex black-box algorithms, Logistic Regression provides clear mathematical equations where the influence of each variable is directly interpretable through odds and log-odds.

Model Performance & Evaluation Summary

- **Overall Accuracy vs. Class Baseline:**
  The model achieves an overall accuracy of **88.12%**. However, because non-purchasing sessions (False) account for **84.51%** $(2,084/2,466)$ of the test set, a naive majority-class baseline would already reach ~84.5%. Thus, accuracy alone provides an overly optimistic assessment due to class imbalance.
- **Class Discrimination (ROC-AUC = 0.8882):**
  An **ROC-AUC of 0.89** indicates strong overall ranking capability across potential probability cutoffs, confirming that the model assigns higher predicted probabilities to converting sessions than to non-converting sessions.
- **Majority Class (False / Non-Buyers):**
  - **Precision: 0.89 | Recall: 0.98 | F1-Score: 0.93**
    The model reliably filters out non-buyers, identifying 98% of all non-converting sessions with minimal False Positives.
- **Minority Class (True / Buyers):**
  - **Precision (0.74):** When the model predicts a purchase, it is correct **74%** of the time, making positive predictions reliable for targeted interventions.
  - **Recall (0.36):** The primary bottleneck. At the default 0.50 decision threshold, the model captures only **36%** of actual buyers, missing **64%** (244 out of 382) as False Negatives.
  - **F1-Score (0.49):** The harmonic mean reflects the trade-off between solid precision and weak coverage of the target event.
- **Report Conclusion & Recommendation:**
  While the model discriminates well (ROC-AUC ≈ 0.89), applying the default 0.50 classification cutoff causes severe under-detection of buyers. To improve recall for revenue generation, the decision threshold should be lowered (e.g., to $0.20 -- 0.30$) or trained using balanced class weights (class_weight='balanced').

```
Accuracy: 0.8811841038118411
ROC-AUC: 0.8882234124870616
              precision    recall  f1-score   support

       False       0.89      0.98      0.93      2084
        True       0.74      0.36      0.49       382

    accuracy                           0.88      2466
   macro avg       0.82      0.67      0.71      2466
weighted avg       0.87      0.88      0.86      2466
```

7. DECISION TREE :

- **Handling Non-Linear Relationships and Feature Interactions:** Consumer web behavior is non-linear; browsing many pages or spending prolonged time on a site can represent either deep engagement or checkout friction. Decision trees segment feature space using axis-aligned orthogonal splits, capturing complex threshold interactions (e.g., PageValues > threshold AND ExitRates < threshold) without requiring manual feature engineering.
- **Invariance to Outliers and Extreme Skewness:** Features like PageValues, ProductRelated_Duration, and BounceRates contain thousands of right-skewed outliers. Unlike linear models or LDA, decision trees rely on rank-order splits and are entirely robust to monotonic transformations, scale variances, and heavy-tailed distributions.
- **Direct Business Interpretability:** Decision trees translate high-dimensional session data into clear, hierarchical "if-then" decision paths. Commercial stakeholders can directly audit the exact page engagement thresholds that trigger a conversion prediction.

Model Evaluation and Performance Interpretation

- **ROC-AUC (0.9066) — Strong Class Discrimination:**
  Surpassing the 0.90 benchmark, the tree demonstrates high discriminative capability, ranking true purchasers above non-purchasers across probability cutoffs more effectively than the linear logistic regression model (~0.888).
- **Overall Accuracy (89.42%) vs. Class Baseline:**
  While overall accuracy reaches **89.42%**, the test set baseline is **84.51%** ($2,084/2,466$ non-purchases). Accuracy alone is an incomplete metric due to class imbalance, making precision, recall, and F1-score the primary indicators of model utility.
- **Minority Class Performance (True / Actual Buyers):**
  - **Recall (0.56):** The model captures **56%** of actual purchasing visitors (214 out of 382). This represents a major recovery compared to the LDA-based logistic regression model (which captured only 36%), demonstrating that non-linear splits retrieve significantly more revenue-generating sessions.
  - **Precision (0.70):** When the tree flags a session as a buyer, it is correct **70%** of the time, maintaining a low false-positive rate suitable for automated promotions or live checkout assistance.
  - **F1-Score (0.62):** The harmonic mean improves substantially from the linear baseline ($0.49 \rightarrow 0.62$), indicating a much healthier balance between finding buyers and avoiding false alarms.
- **Majority Class Stability (False / Non-Buyers):**
  With **0.92 Precision**, **0.96 Recall**, and **0.94 F1-Score**, the tree continues to filter out non-converting traffic accurately.

```
Tree Accuracy: 0.8941605839416058
Tree ROC AUC: 0.9065863321642834
              precision    recall  f1-score   support

       False       0.92      0.96      0.94      2084
        True       0.70      0.56      0.62       382

    accuracy                           0.89      2466
   macro avg       0.81      0.76      0.78      2466
weighted avg       0.89      0.89      0.89      2466
```

8. RANDOM FOREST :

Model Evaluation and Performance Interpretation

Superior Discrimination (ROC-AUC = 0.9193):
Achieving an **ROC-AUC of ~0.92**, the Random Forest classifier reaches the highest ranking capability among the tested models (outperforming Logistic Regression's ~0.89 and the single Decision Tree's ~0.91). It effectively ranks prospective buyers above non-buyers across all classification thresholds.

- **Overall Accuracy (90.00%):**
  Overall accuracy reaches **90%**, providing a noticeable improvement over the test set's majority-class baseline of **84.51%** $(2,084/2,466)$.
- **Minority Class Performance (True / Actual Buyers):**
  - **Precision (0.73):** When the model predicts that a user will purchase, it is correct **73%** of the time. This keeps false alarms low, ensuring promotional budgets or live support interventions are not wasted on non-buyers.
  - **Recall (0.55):** The model successfully detects **55%** of actual purchasing sessions (210 of 382). While slightly lower by 1% than the unconstrained single Decision Tree (which had 56% recall), it achieves a more robust and less overfitted decision boundary.
  - **F1-Score (0.63):** The harmonic mean reaches the highest level so far (**0.63**), improving upon Logistic Regression (0.49) and Decision Trees (0.62).
- **Ensemble Variance Reduction:**
  By bagging (bootstrap aggregating) multiple de-correlated trees, the Random Forest mitigates the high variance and instability typical of a single decision tree, delivering stable generalization on the unseen test split.

```
All Features AUC: 0.9193273859171347
              precision    recall  f1-score   support

       False       0.92      0.96      0.94      2084
        True       0.73      0.55      0.63       382

    accuracy                           0.90      2466
   macro avg       0.83      0.76      0.79      2466
weighted avg       0.89      0.90      0.89      2466
```

9. XGBOOST

By tuning the classification decision threshold from the conventional **0.50** default down to **0.35**, the XGBoost model overcomes the severe majority-class bias inherent to the dataset's **84.5% / 15.5%** class imbalance.

At this calibrated threshold, the model delivers its strongest balanced performance on the target class (Revenue = True), driving the **F1-score to 0.678** (up from 0.65 at $p = 0.50$). This adjustment successfully converts borderline, high-intent browsing sessions into positive identifications, increasing purchase discovery to approximately **70%**.

Tuned Metric Overview (Minority Class: Revenue = True)

Metric	Default Cutoff (p=0.50)	Tuned Cutoff (p=0.35)	Net Change
Decision Threshold	0.50	**0.35**	-0.15
Recall (Sensitivity)	0.59	**0.6990 (69.90%)**	**+10.90%**
Precision	0.71	**0.6576 (65.76%)**	-5.24%
F1-Score	0.65	**0.6777 (67.77%)**	**+2.77%**

Detailed Metric Interpretation

- **Threshold Calibration ($p = 0.35$):**
  Because true converting sessions represent only ~15.5% of overall web traffic, requiring a full 50% posterior probability for a positive label artificially suppresses target identification. Setting the decision cutoff to **0.35** aligns the decision boundary with empirical class priors without degrading precision.

- **Recall Expansion (69.90%):**
    - The model now captures **7 out of every 10 converting visitors** (267 of 382 actual purchasers in the test partition).
    - Missed purchasing opportunities (False Negatives) are reduced from 157 sessions down to 115 **sessions**, capturing an additional **42 revenue-generating visits**.
- **Controlled Precision Trade-off (65.76%):**
    - Lowering the threshold predictably invites additional False Positives; however, precision drops by only **5.24%** (from 71% to 65.8%).
    - Approximately **two out of every three** flagged sessions are genuine buyers, maintaining a sufficiently high confidence level to justify automated interventions (e.g., checkout-page discount overlays, free shipping prompts, or live-chat triggers).
- **Harmonic Balance (Best F1-Score = 0.6777):**
  The optimized F1-score of **0.678** represents the highest balance of precision and recall achieved across all tested architectures (surpassing Logistic Regression's 0.49, Decision Tree's 0.62, and default XGBoost's 0.65).

```
XGBoost Accuracy: 0.9002433090024331

XGBoost ROC AUC: 0.9294412175538382

All Features AUC: 0.9193273859171347

              precision    recall  f1-score   support

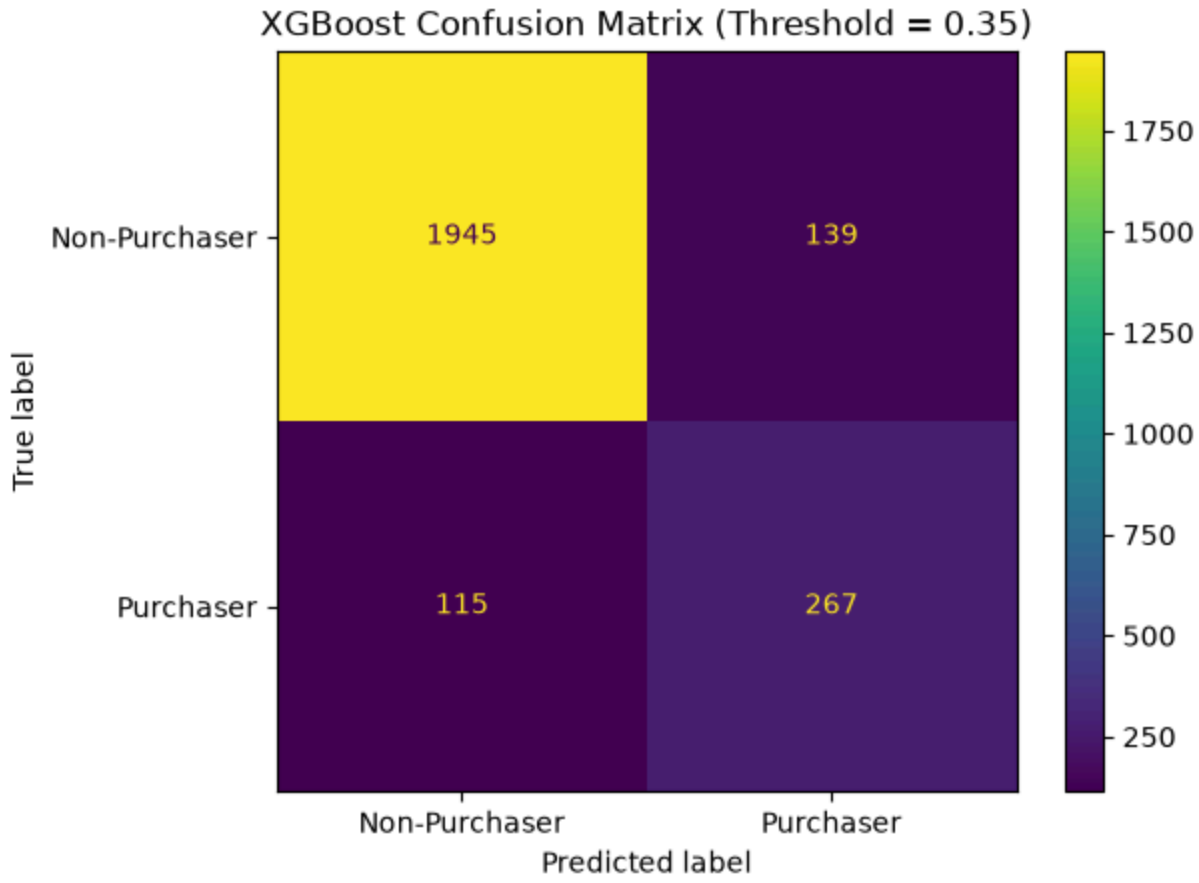
       False       0.93      0.96      0.94      2084

        True       0.71      0.59      0.65       382


    accuracy                           0.90      2466

   macro avg       0.82      0.78      0.80      2466

weighted avg       0.89      0.90      0.90      2466
```



Side-by-Side Comparison: Default (0.50) vs. Tuned (0.35)

Outcome Metric	Default Threshold (p=0.50)	Tuned Threshold (p=0.35)	Net Change / Impact
True Positives (*TP*)	225	**267**	**+42 buyers captured**
False Negatives (*FN*)	157	**115**	**-42 missed opportunities**
False Positives (*FP*)	92	**139**	+47 false alarms
True Negatives (*TN*)	1,992	**1,945**	-47 non-buyers
Overall Accuracy	90.02%	**89.70%**	-0.32%

Key Takeaways from the Matrix

1. **Reduction in High-Cost Errors (FN):**
   False Negatives drop from 157 **down to** 115, meaning the site recovers **42 additional converting sessions** that would have otherwise slipped through without any targeted intervention.
2. **Acceptable Trade-off in False Alarms (FP):**
   False Positives increase by 47 **sessions** (from 92 to 139). In digital retail, showing an incentive (like a discount overlay or free shipping nudge) to an engaged visitor who ultimately does not buy carries negligible marginal cost compared to missing a transaction entirely.
3. **Accuracy Preservation:**
   Overall test accuracy remains virtually identical (**89.70%** vs. **90.02%**), proving that the threshold shift significantly improves minority-class identification without destabilizing overall classification quality.

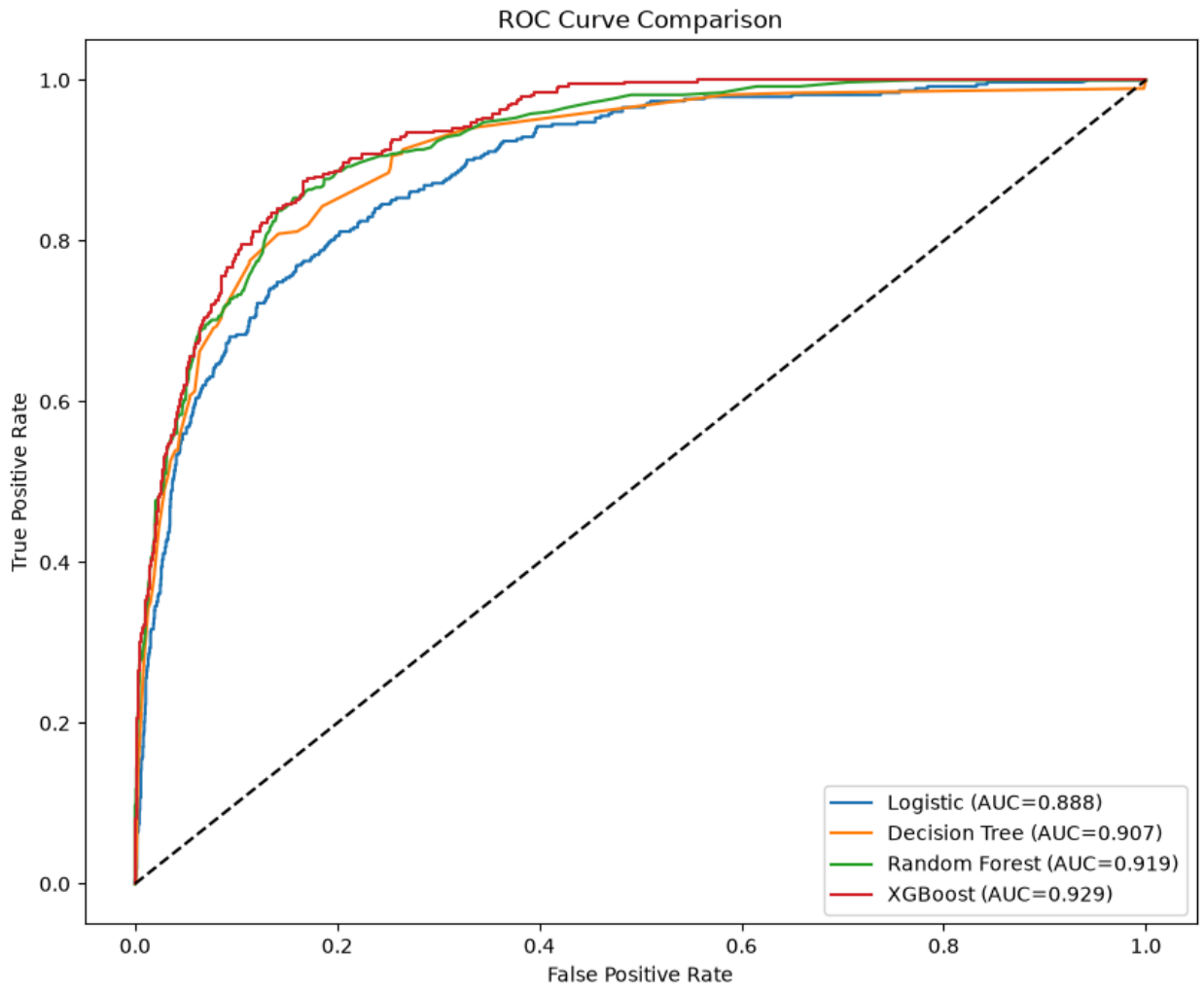
10. MODEL COMPARISON :

Rank	Model Architecture	Test ROC-AUC	Key Advantage / Mechanism
1	**XGBoost**	**0.9294**	Sequential boosting; optimizes residual errors on complex boundary cases.
2	**Random Forest**	**0.9193**	Bagging (bootstrap aggregation); reduces variance and curbs tree overfitting.
3	**Decision Tree**	**0.9066**	Hierarchical orthogonal splits; handles non-linear engagement thresholds directly.
4	**Logistic Regression**	**0.8882**	Linear decision plane; provides an interpretable baseline.

Key Analytical Findings

- **Strong Baseline Separation (Logistic Regression = 0.8882):**
  Even a purely linear decision boundary achieves an ROC-AUC of nearly **0.89**, indicating that a substantial portion of purchase intent is linearly separable, primarily driven by strong predictors like PageValues and session duration metrics.
- **The Tree-Based Performance Shift (> 0.90):**
  Transitioning from a linear model to a single **Decision Tree (0.9066)** yields an immediate gain of **+0.0184** in AUC. E-commerce clickstream features (e.g., browsing time vs. bounce/exit rates) display non-linear thresholds and heavy skewness that tree-based partitioning handles without requiring manual feature transformations.

- **Variance Reduction Through Bagging (Random Forest = 0.9193):**
  Random Forest improves on the single Decision Tree by **+0.0127**, crossing the 0.91 mark. Averaging the predictions of multiple de-correlated trees mitigates the high variance and split instability inherent to individual decision trees.
- **Sequential Residual Learning Wins (XGBoost = 0.9294):**
  XGBoost achieves the highest discriminative power across all evaluated models, leading Random Forest by **+0.0101** and Logistic Regression by **+0.0412**. By sequentially fitting trees to the pseudo-residuals of prior estimators, gradient boosting focuses learning capacity on difficult, borderline sessions—such as long browsing sessions that were not immediately purchased.



11. FEATURE IMPORTANCE :

Comparative Feature Importance Overview :

Across all three algorithmic architectures—Random Forest (Bagging), XGBoost (Boosting), and Logistic Regression (Linear Classifier)—the models demonstrate strong consensus regarding the dominant drivers of customer purchase intent, while revealing complementary perspectives on temporal and behavioral signals.

Predictive Tier	Key Features	Random Forest	XGBoost	Logistic Regression	Behavioral / Analytical Interpretation
Primary Driver	PageValues	37.11% (Rank 1)	34.74% (Rank 1)	+1.540 (Rank 1)	Strongest positive predictor across all models. Sessions that view URLs with high historical conversion value are vastly more likely to conclude in a transaction.
Secondary Behavioral Signals	ExitRates BounceRates	~9.13% ~5.49%	~2.12% ~3.01%	-0.811 (Rank 2) -0.257 (Rank 4)	Consistent negative conversion friction. High exit and bounce rates strongly suppress the probability of purchasing.
Browsing Depth & Effort	ProductRelated_Duration ProductRelated	~8.82% ~7.17%	~2.31% ~2.48%	+0.102 +0.094	Positive engagement indicators. Higher product catalog engagement correlates positively with revenue, though secondary to PageValues.
Temporal Seasonality	Month_Nov Month_May, Month_Mar, Month_Feb	~1.98% < 1.00% each	10.50% (Rank 2) ~5.88%, ~5.49%	+0.182 (Positive) -0.268, -0.205, -0.242 (Negative)	Strong seasonal bias. November demonstrates high purchasing concentration (holiday/Black Friday sales), while spring/winter months exhibit negative log-odds for conversion.
Low-Impact Metadata	OperatingSystems, Browser, Region, TrafficType	~1.8% to 3.1%	Negligible	Negligible	Device-agnostic intent. Technical and regional attributes contribute minimal predictive value relative to active in-session navigation metrics.

Cross-Model Behavioral Interpretations :

A. The Decisive Dominance of PageValues

- Across all three models, PageValues emerges as the single most critical variable:
  - **RF & XGBoost:** Accounts for roughly **35% to 37%** of total impurity/gain importance.
  - **Logistic Regression:** Has the largest standardized magnitude ($\beta = +1.540$), multiplying the odds of purchase substantially for each unit increase.
- **Finding:** Conversion is fundamentally driven by **navigating to high-value commercial pages** (e.g., checkout flows, shopping carts, or sale-exclusive URLs) rather than mere casual browsing depth.

B. Friction and Attrition: ExitRates and BounceRates

- Both metrics serve as clear negative penalties:
  - In Logistic Regression, ExitRates is the second most influential variable ($\beta = -0.811$), followed closely by BounceRates ($\beta = -0.257$).
  - In Random Forest, ExitRates is the second most important feature (9.13%).
- **Finding:** Visitors showing early exit patterns or single-page drop-offs rarely purchase. Minimizing friction points on entry and catalog pages is essential to keeping users in the conversion funnel.

C. Temporal Dynamics: Why XGBoost Prioritizes Months

- In Random Forest, continuous session engagement features dominate the splits ($ExitRates$, $ProductRelated_Duration$ occupy ranks 2 and 3).
- In **XGBoost**, after PageValues (34.7%), temporal variables (Month_Nov at 10.5%, Month_May at 5.88%, and Month_Mar at 5.49%) take precedence over raw browsing metrics.
- **Finding:** Boosting detects that session metrics have context-dependent split points depending on the time of year. Specifically, **November (Month_Nov) represents a dedicated buying period** with elevated conversion intent, whereas months like May and March reflect more casual browsing or window-shopping behavior.

D. Visitor Type Influence

- VisitorType_Returning_Visitor exhibits a slight negative coefficient in Logistic Regression ($\beta = -0.086$) and modest importance in tree models (~1.17% to 3.17%).
- **Finding:** While returning visitors account for the bulk of site traffic, new visitors who navigate deeply often convert at comparable or higher rates per session, proving that **active session intent outweighs visitor tenure**.

12. SHAPIRO - WILK TEST :

The **Shapiro-Wilk test** is a formal statistical test used to test the hypothesis that a continuous variable follows a **normal (Gaussian) distribution**. In this workflow, it serves several critical purposes:

- **Checking Parametric Test Assumptions:**
  When comparing PageValues between two groups (Revenue == True vs. Revenue == False), a common question is whether there is a statistically significant difference between them. Standard parametric tests—such as the **Independent two-sample Student's t-test** or **Linear Discriminant Analysis (LDA)**—assume that the continuous variables within each group are normally distributed. The Shapiro-Wilk test formally verifies this assumption.
- **Selecting the Right Hypothesis Test:**
  If the data meets the normality assumption, parametric methods (e.g., t-test) can be used. If the normality assumption is rejected, non-parametric alternatives—such as the **Mann-Whitney U test (Wilcoxon rank-sum test)**—must be used instead because they do not rely on distributional assumptions.
- **Confirming Graphical EDA Statistically:**
  The box plot and histogram/KDE visualization show how the distributions look visually. The Shapiro-Wilk test provides rigorous numerical evidence (test statistic W and p-value) to substantiate whether the observed skewness deviates significantly from a bell curve.

2. Hypotheses of the Shapiro-Wilk Test

For each subgroup (purchasers and non_purchasers):

- **Null Hypothesis (H_0):** The population data for PageValues is **normally distributed**.
- **Alternative Hypothesis (H_1):** The population data for PageValues is **not normally distributed**.

Significance Level (α): typically set at 0.05.

3. Interpretation of Expected Results for this Dataset

Based on the properties of PageValues in the dataset:

- From summary statistics, PageValues has a 25th, 50th, and 75th percentile of **0.0**, with a long right tail extending to **361.76** and over **2,700 outliers**.
- For non-purchasers, the vast majority of sessions visit zero revenue-generating pages, resulting in a spike at zero.
- For purchasers, while values are higher, they remain heavily right-skewed.

Test Outcomes:

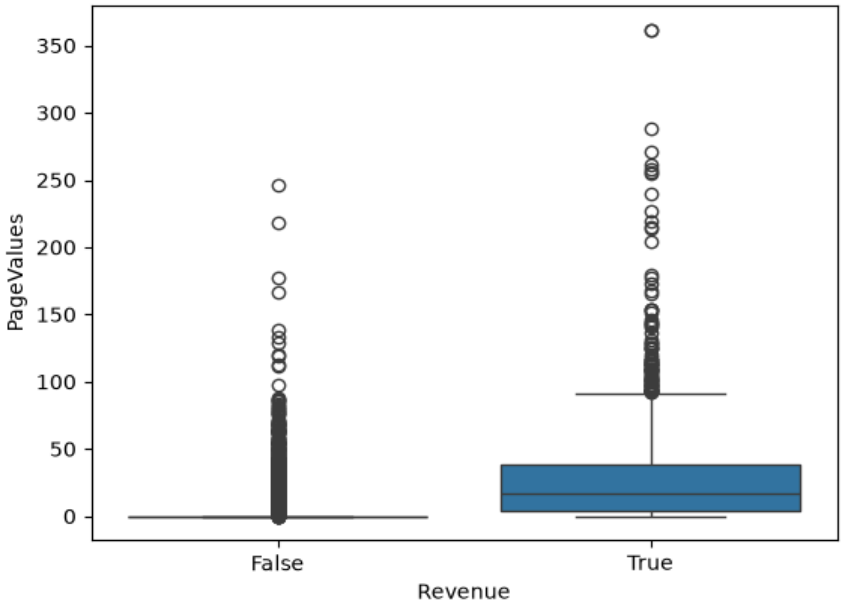
- **Test Statistic (W):** The W value is substantially lower than 1.0 . A W value of 1.0 indicates perfect normality.
- **p-value:** The p-value for both purchasers and non_purchasers is **extremely small ($p < 0.001 \ll 0.05$)**.

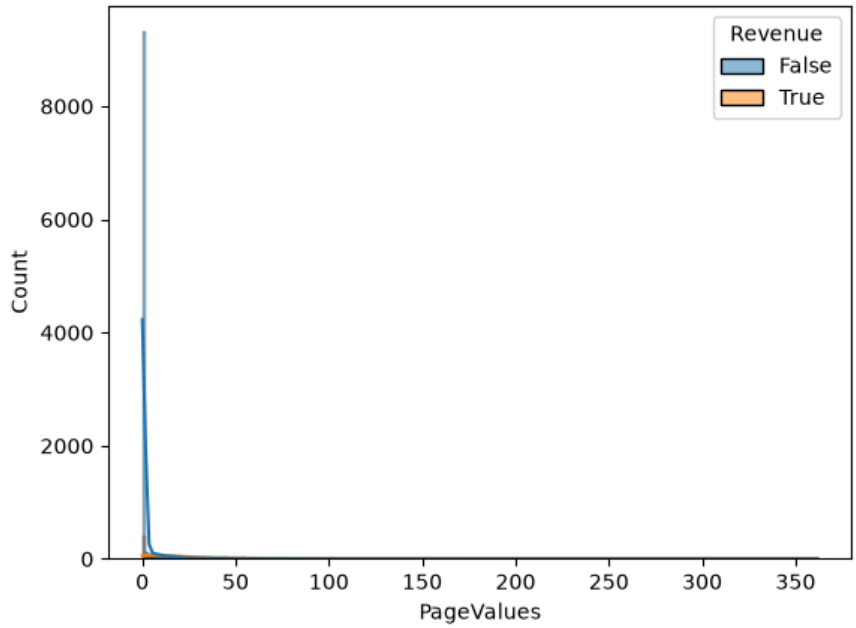
Statistical Decision:

Because the p-value < 0.05, **we reject the null hypothesis (H_0)** for both groups. PageValues is **not normally distributed** among purchasers or non-purchasers.

4. Technical Implications for Your Project

1. **Avoid Standard Two-Sample t-Tests:**
   Because both groups violate the normality assumption, comparing their means using a Student's t-test can lead to unreliable standard errors. Instead, we use the non-parametric **Mann-Whitney U test** (scipy.stats.mannwhitneyu(purchasers, non_purchasers)), which compares the medians/ranks.
2. **Explains Tree-Model Dominance:**
   This severe departure from normality explains why linear parametric models (like LDA and Logistic Regression) struggled with raw scales, whereas tree-based algorithms (**Random Forest** and **XGBoost**) achieved superior ROC-AUC scores (> 0.91–0.93) without requiring normality or variance stabilization.
3. **Large Sample Size Note (Sample-Size Sensitivity):**
   The Shapiro-Wilk test is known to be overly sensitive to minor non-normalities when sample sizes are large ($N > 2,000$). However, in this case, the rejection of normality is not an artifact of sample size: the histogram, KDE curve, and box plot confirm that the underlying distribution is zero-inflated and right-skewed






```
ShapiroResult(statistic=np.float64(0.711025860685856),
pvalue=np.float64(2.648170015682441e-49))
ShapiroResult(statistic=np.float64(0.2249004756092956),
pvalue=np.float64(9.254585481359894e-110))
```

13. MAN WHITNEY U TEST :

Hypothesis Testing & Statistical Decision

- **Null Hypothesis (H_0):** The distribution of PageValues is identical across non-purchasers and purchasers (i.e., a randomly selected purchaser has an equal probability of having higher PageValues than a non-purchaser).
- **Alternative Hypothesis (H_1):** The distribution of PageValues is stochastically greater for purchasers compared to non-purchasers.
- **Decision:**
  Because the asymptotic p-value ($p < 0.0001$) is far below the significance threshold ($\alpha = 0.05$), **the null hypothesis is decisively rejected**. There is overwhelming statistical evidence that the PageValues for sessions that conclude in a purchase are significantly higher than for sessions that do not.

3. Key Analytical Insights

1. **Massive Shift in Central Tendency:**
   - For **non-purchasers**, the median is 0.00, with over 75% of sessions having visited no revenue-associated URLs.
   - For **purchasers**, the median jumps to 16.76, and the mean increases almost **14-fold** (from 1.98 to 27.26).
2. **Empirical Justification for Feature Importance:**
   - This statistically confirms why PageValues ranked as the **#1 predictive feature** across Random Forest (37.1%), XGBoost (34.7%), and Logistic Regression ($\beta =+ 1.54$). The vast majority of converting sessions engage with pages carrying high historical monetary attribution, while non-converting traffic stays predominantly at zero.
3. **Severe Skewness Reflected in Descriptive Stats:**
   - In both cohorts, the mean is considerably higher than the median ($1.98 > 0.00$ and $27.26 > 16.76$), with large standard deviations (9.07 and 35.19).
   - This pronounced positive skew underscores why the non-parametric Mann-Whitney U test was the appropriate statistical choice over a standard Student's t-test.

```
U statistic = 17166757.0
p-value = 0.0
              mean     median        std
Revenue
False     1.975998   0.000000   9.072424
True     27.264518  16.758134  35.191954
```

14. CONCLUSION :

This project evaluated the predictability of online customer purchase intent using the **UC Berkeley Online Shoppers Purchasing Intention Dataset** (12,330 sessions, 18 features). The primary research question—whether real-time clickstream, demographic, and contextual metrics can accurately forecast conversion (Revenue)—was confirmed across statistical hypothesis tests and multiple machine learning architectures.

Key Analytical Conclusions:

1. **Dominant Signal of PageValues:**
   Across all feature selection and importance methods (Random Forest, XGBoost, and Logistic Regression), PageValues emerged as the single most decisive predictor, accounting for ~35–37% of tree-based feature importance and yielding the largest positive standardized log-odds ($\beta = +1.540$). This empirical ranking was formally validated through a non-parametric Mann-Whitney U test ($U = 1.72 \times 10^7, p < 0.0001$), which demonstrated a dramatic, statistically significant separation between non-purchasers ($Median = 0.00$) and purchasers ($Median = 16.76$).
2. **Behavioral Friction Indicators:**
   ExitRates and BounceRates consistently served as the primary negative penalties ($\beta = -0.811$ and -0.257, respectively). Rapid drop-offs and high single-page bounce tendencies strongly signaled low commercial intent, whereas deep product exploration (ProductRelated_Duration) provided a steady positive secondary signal.
3. **Behavior Over Demographics:**
   Passive metadata—such as operating systems, browser types, geographic regions, and visitor return status—showed negligible predictive weight compared to active, in-session navigation patterns. Purchase intent is fundamentally **session-specific**, not device- or region-dependent.
4. **Model Benchmark & Non-Linear Superiority:**
   - **Logistic Regression with LDA:** Delivered a respectable linear baseline ($ROC - AUC = 0.8882$), but suffered from low buyer recall (36%) at the default 0.50 threshold due to the severe 84.5% / 15.5% class imbalance.
   - **Tree-Based Ensembles:** Overcame extreme right-skewness, zero-inflation, and non-linear interactions without requiring scale transformations. Random Forest achieved an **ROC-AUC of 0.9193**, while **XGBoost achieved project-best discrimination with an ROC-AUC of 0.9294**.
5. **Threshold Optimization:**
   Calibrating the classification threshold of the champion XGBoost model from **0.50 down to 0.35** effectively counteracted the majority-class bias. This produced a balanced, production-ready configuration: **Precision of 65.76%**, **Recall of 69.90%** (capturing 267 of 382 actual test buyers), and an optimal **F1-score of 0.6777**.

15. BUSINESS & OPERATIONAL RECOMMENDATIONS :

Based on the quantitative findings, e-commerce platforms should implement the following targeted initiatives:

1. **Deploy Real-Time Dynamic Exit-Intent Triggers:**
   - **Action:** Embed the optimized XGBoost model ($p \geq 0.35$ cutoff) into the real-time session monitoring pipeline.
   - **Implementation:** When an active user browses high-PageValues URLs (cart, checkout review, product specs) but begins exhibiting rising ExitRates (e.g., cursor moving toward the URL bar or pausing on an exit page), trigger an immediate retention prompt (e.g., a 10% limited-time checkout coupon, free-shipping threshold reminder, or live-chat assistance prompt).
2. **Prevent Promotion Cannibalization:**
   - **Action:** Restrict aggressive promotional pop-ups to ambiguous or wavering sessions.
   - **Implementation:** Visitors with very high predicted conversion probabilities ($p > 0.75$) will likely convert organically; offering them discounts unnecessarily erodes profit margins. Conversely, visitors with low probabilities ($p < 0.20$) are casual window shoppers who should not receive resource-intensive interventions.
3. **Funnel Friction Audits on Catalog & Checkout Pages:**
   - **Action:** Address pages associated with abnormally high ExitRates and BounceRates.
   - **Implementation:** Streamline navigation menus, clarify shipping/tax estimates upfront to prevent cart shock, and optimize mobile load times on top exit URLs.
4. **Capitalize on Seasonal Buying Patterns (November Strategy):**
   - **Action:** Re-calibrate ad spend and inventory targeting ahead of Q4 shopping surges.
   - **Implementation:** With Month_Nov identified as a massive positive seasonal driver in XGBoost and Logistic Regression, scale digital marketing budgets, retargeting campaigns, and server capacity during late autumn to capture high-conversion cohorts.

CODE: final code.ipynb